

This script computes the metrics used in the final text and reporting recipe

```
import numpy as np
import pandas as pd
import geopandas as gpd
import os.path as path
import utils
import matplotlib.pyplot as plt
from collections import Counter

# load all files previously computed
lau_df = utils.load_complex_geojson(
    path.join(utils.out_data_dir, "drought_days_lau.geojson")
)
nuts3_df = utils.load_complex_geojson(
    path.join(utils.out_data_dir, "nuts3_stats.geojson")
)
country_df = pd.read_json(path.join(utils.out_data_dir, "country_stats.geojson"))
```

Most drought-stricken countries

This outputs all analyzed countries in order, based on how many drought days they experience on average per year.

```
for i, cntr in enumerate(
    country_df.sort_values(by="median_drought_days", ascending=False)[
        "NAME_ENGL"
    ].values
):
    print(i + 1, cntr)
```

- 1 Albania
- 2 Serbia
- 3 Poland
- 4 Liechtenstein
- 5 North Macedonia
- 6 Bulgaria
- 7 Greece
- 8 Lithuania
- 9 Switzerland
- 10 Romania
- 11 France
- 12 Croatia
- 13 Czechia
- 14 Slovenia
- 15 Slovakia
- 16 Spain
- 17 Italy
- 18 Cyprus
- 19 Luxembourg
- 20 United Kingdom
- 21 Germany
- 22 Hungary
- 23 Belgium
- 24 Netherlands
- 25 Ireland
- 26 Austria
- 27 Malta
- 28 Portugal
- 29 Denmark
- 30 Sweden
- 31 Finland
- 32 Iceland
- 33 Estonia
- 34 Norway
- 35 Latvia

```
country_df.sort_values(by="max_drought_days")[:5][  
    ["NAME_ENGL", "median_drought_days", "max_drought_days"]  
]
```

	NAME_ENGL	median_drought_days	max_drought_days
NO	Norway	34.11	69.70
FI	Finland	40.98	104.12
AT	Austria	56.52	110.29
SE	Sweden	44.94	119.68
IS	Iceland	35.93	148.81

Looking at the countries with the lowest “max_drought_days” value: -> all countries in Europe have faced drought in the observed timeframe.

Europe-wide statistics

This section splits the nuts3-dataset along a threshold for median and maximum drought days. From this it computes a portion of surface area

```
drought_days_median_50_mask = nuts3_df["median_drought_days"] > 50
dry_nuts3_df = nuts3_df[drought_days_median_50_mask]
wet_nuts3_df = nuts3_df[~drought_days_median_50_mask]
dryland_km2, wetland_km2 = (
    dry_nuts3_df["area_km2"].sum(),
    wet_nuts3_df["area_km2"].sum(),
)
dryland_km2 / (dryland_km2 + wetland_km2)
```

```
np.float64(0.6541255404617256)
```

-> almost two thirds of the EEA+UK area experiences on average more than 50 days of drought warnings per year.

```
drought_days_median_100_mask = nuts3_df["median_drought_days"] > 100
dry_nuts3_df = nuts3_df[drought_days_median_100_mask]
wet_nuts3_df = nuts3_df[~drought_days_median_100_mask]
dryland_km2, wetland_km2 = (
    dry_nuts3_df["area_km2"].sum(),
    wet_nuts3_df["area_km2"].sum(),
)
dryland_km2 / (dryland_km2 + wetland_km2)
```

```
np.float64(0.11943094078060766)
```

-> Over 10% of the EEA+UK area experiences on average more than 100 days of drought warnings per year.

```
drought_days_max_183_mask = nuts3_df["max_drought_days"] >= 183
dry_nuts3_df = nuts3_df[drought_days_max_183_mask]
wet_nuts3_df = nuts3_df[~drought_days_max_183_mask]
dryland_km2, wetland_km2 = (
    dry_nuts3_df["area_km2"].sum(),
    wet_nuts3_df["area_km2"].sum(),
)
dryland_km2 / (dryland_km2 + wetland_km2)
```

```
np.float64(0.6648711964996835)
```

-> Two thirds of all regions in EEA+UK have experienced a severe drought year since 2012, in which a drought warning was issued for over 183 Days (half a year).

NUTS-3-regions with the highest median number of drought days

```
nuts3_df.sort_values(by="median_drought_days", ascending=False)[:10][
    ["nuts_id", "cntr_code", "name_latn", "median_drought_days", "max_drought_days"]
]
```

	nuts_id	cntr_code	name_latn	median_drought_days	max_drought_days
8	AL032	AL	Fier	199.85	334.84
1110	PL923	PL	Płocki	178.71	345.02
1099	PL822	PL	Przemyski	175.58	253.82
128	CH032	CH	Basel-Landschaft	167.22	281.20
670	ES611	ES	Almería	165.54	307.81
1279	UKE32	UK	Sheffield	165.23	279.97
11	AL035	AL	Vlorë	162.00	267.75
609	EL542	EL	Thesprotia	159.92	309.65
983	MK005	MK	Pelagoniski	156.90	306.41
607	EL533	EL	Florina	156.30	273.65

```
lau_df.sort_values(by="median_drought_days", ascending=False)[:10][
    ["GISCO_ID", "CNTR_CODE", "LAU_NAME", "median_drought_days", "max_drought_days"]
]
```

	GISCO_ID	CNTR_CODE	LAU_NAME	median_drought_days	max_drought_d
701	ES_18056	ES	Cúllar	280.94	357.73
18662	EL_41010301	EL	K A T	275.00	340.00
18670	EL_41010309	EL	K II	275.00	340.00
2237	ES_18029	ES	Benamaurel	270.95	359.75
2742	ES_18187	ES	Villanueva de las Torres	263.94	360.00
18665	EL_41010304	EL	K A	262.90	349.54
18669	EL_41010308	EL	K II	252.42	346.28
707	ES_18064	ES	Dehesas de Guadix	250.94	352.96
15048	EL_10020117	EL	K T	248.38	314.28
18666	EL_41010305	EL	K H	246.38	348.17

Note: These LAU stats are pretty juicy, but they are super inconsistent - Germany has around 400 LAUs, while Cyprus has only one, which massively skews results. So we don't use these

```
avg_drought_days_gt_100 = len(lau_df[lau_df["median_drought_days"] > 100])
print(
    f"{avg_drought_days_gt_100} of {len(lau_df)} LAUs ({100* avg_drought_days_gt_100 / len(lau_df)}%) experience more than 100 drought days per year.
```

19362 of 99114 LAUs (19.54%) experience more than 100 drought days per year.

```
max_drought_days_gt_200 = len(lau_df[lau_df["max_drought_days"] > 200])
print(
    f"{max_drought_days_gt_200} of {len(lau_df)} LAUs ({100* max_drought_days_gt_200 / len(lau_df)}%) experience more than 200 drought days per year.
```

```
print("most frequent 'worst year' at LAU-scale:")
print(lau_df["max_drought_days_year"].value_counts()[:5])
```

82779 of 99114 LAUs (83.52%) experienced at least one year between 2012 and 2025 with over 200 drought days per year.

most frequent 'worst year' at LAU-scale:

max_drought_days_year

2019	20158
2022	19760
2017	13365

```
2012    13045
2018     7216
Name: count, dtype: int64
```